

CDAC Mumbai

Lab Assignment 2

Problem 1: Grade Evaluation System

Problem Statement:

Write a Java program that calculates the average marks of a student and determines the grade based on the following criteria:

Grade A: Average marks ≥ 90

Grade B: Average marks between 70 and 89

Grade C: Average marks between 50 and 69

Grade D: Average marks between 30 and 49

Fail: Average marks < 30

Predefined Values (Try with different values as well):

- Marks in Maths = 80
- Marks in Science = 85
- Marks in History = 90

Expected Output:

Average Marks: 85

Grade: B Problem

Java Code:

```
public class Grade {  
  
    public static void main(String[] args) {  
  
        int maths = 96;  
        int science = 92;  
        int history = 90;  
  
        int average = (maths+science+history)/3;  
  
        System.out.println("Average Marks: " + average);  
  
        if (average >=90) {  
            System.out.println("Grade A");  
        }  
        else if (average >=70) {
```

```

        System.out.println("Grade B");
    }
    else if (average >=50) {
        System.out.println("Grade C");
    }
    else if (average>=30) {
        System.out.println("Grade D");
    }
    else {
        System.out.println("Fail");
    }
}
}

```

Output:

PS F:\Java Code\Logic Building\Assignment 2> javac Grade.java

PS F:\Java Code\Logic Building\Assignment 2> java Grade

Average Marks: 85

Grade B

➤ Output with different values as well:

1. Values: (Maths=10, Science=9, History=17)

PS F:\Java Code\Logic Building\Assignment 2> javac Grade.java

PS F:\Java Code\Logic Building\Assignment 2> java Grade

Average Marks: 12

Fail

2. Values: (Maths=96, Science=93, History=90)

PS F:\Java Code\Logic Building\Assignment 2> javac Grade.java

PS F:\Java Code\Logic Building\Assignment 2> java Grade

Average Marks: 93

Grade A

Problem 2: Leap Year

Problem Statement:

Write a Java program that checks whether the year is a leap year or not. A year is a leap year if: It is divisible by 4, but not divisible by 100, or It is divisible by 400.

Predefined Value (Try with different values as well):

- Year = 2024
- Year = 1900

Expected Output:

2024 is a leap year. 1900 is not a leap year.

Java Code:

```
public class LeapYear {  
    public static void main(String[] args) {  
        int year = 1968;  
        if (year % 400 == 0 || year % 4 == 0 && year % 100 != 0) {  
            System.out.println(year + " is a leap Year");  
        } else {  
            System.out.println(year + " is not a leap Year");  
        }  
    }  
}
```

Output:

```
PS F:\Java Code\Logic Building\Assignment 2> javac LeapYear.java  
PS F:\Java Code\Logic Building\Assignment 2> java LeapYear  
2024 is a leap Year  
PS F:\Java Code\Logic Building\Assignment 2>
```

➤ Output with different values as well:

1. Year = 1900

```
PS F:\Java Code\Logic Building\Assignment 2> javac LeapYear.java  
PS F:\Java Code\Logic Building\Assignment 2> java LeapYear  
1900 is not a leap Year
```

2. Year = 1968

```
PS F:\Java Code\Logic Building\Assignment 2> javac LeapYear.java  
PS F:\Java Code\Logic Building\Assignment 2> java LeapYear  
1968 is a leap Year  
PS F:\Java Code\Logic Building\Assignment 2>
```

Problem 3: Days of the Week

Problem Statement:

Write a Java program that takes an integer between 1 and 7 and prints the corresponding day of the week using a switch-case statement. If the input is outside the range of 1 to 7, the program should display "Invalid day number".

Predefined Value:

Day number = 3

Expected Output:

The day is Wednesday.

Java Code:

```
public class Day {  
  
    public static void main(String[] args) {  
  
        int day = 3;  
  
        switch (day) {  
            case 1:  
                System.out.println("The day is Monday");  
                break;  
  
            case 2:  
                System.out.println("The day is Tuesday");  
                break;  
  
            case 3:  
                System.out.println("The day is Wednesday");  
                break;  
  
            case 4:  
                System.out.println("The day is Thursday");  
                break;  
  
            case 5:  
                System.out.println("The day is Friday");  
                break;  
  
            default:  
                System.out.println("Invalid day number");  
                break;  
        }  
    }  
}
```

```

        case 6:
            System.out.println("The day is Saturday");
            break;

        case 7:
            System.out.println("The day is Sunday");
            break;

        default:
            System.out.println("Invalid Day Number");

    }
}

```

Output:

```

PS F:\Java Code\Logic Building\Assignment 2> javac Day.java
PS F:\Java Code\Logic Building\Assignment 2> java Day
The day is Wednesday
PS F:\Java Code\Logic Building\Assignment 2>

```

➤ Output with different values as well:

1. Day 10

```

PS F:\Java Code\Logic Building\Assignment 2> javac Day.java
PS F:\Java Code\Logic Building\Assignment 2> java Day
Invalid Day Number
PS F:\Java Code\Logic Building\Assignment 2>

```

2. Day 5

```

PS F:\Java Code\Logic Building\Assignment 2> javac Day.java
PS F:\Java Code\Logic Building\Assignment 2> java Day
The day is Friday
PS F:\Java Code\Logic Building\Assignment 2>

```

Problem 4: Identify the Values of Uninitialized Variables

Scenario:

You are working on a program that handles different data types. Your manager has asked you to quickly check the values of various variables, but you're in a rush and forget to initialize them. As you go through the code, you expect some values to show up, but Java has something else in mind. Your task is to fix the issue and ensure the variables hold proper values.

Instructions:

1. Declare the following variables:

```
byte a;  
short b;  
int c;  
long d;  
float e;  
double f;  
char g;  
boolean h;
```

Java Code:

```
public class Var {  
    public static void main(String[] args) {  
  
        byte a = 0;  
        short b = 0;  
        int c = 0;  
        long d = 0;  
        float e = 0;  
        double f = 0;  
        char g = 'A';  
        boolean h = true;  
  
        System.out.println(a);  
        System.out.println(b);  
        System.out.println(c);  
        System.out.println(d);  
        System.out.println(e);  
        System.out.println(f);  
        System.out.println(g);  
        System.out.println(h);  
  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 2> javac Var.java

PS F:\Java Code\Logic Building\Assignment 2> java Var

0

0

0

0

0.0

0.0

A

true

PS F:\Java Code\Logic Building\Assignment 2>